MetaVG: A Meta-Learning Framework for Visual Grounding

Chao Su, Zhi Li, Tianyi Lei, Dezhong Peng, and Xu Wang

***Abstract*—Visual grounding aims at localizing objects in images using natural language expressions. This task can be challenging when there are significant differences between the distributions of the training and testing sets. Existing methods tend to excessively focus on the training sets, which could lead to overfitting, especially in small-sample scenarios. To address this issue, in this letter, we present a novel meta-learning-based training framework called MetaVG, for visual grounding. Our approach leverages bi-level optimization to adapt quickly to the target task, thereby alleviating the overfitting issue. To train MetaVG effectively, we propose a novel training mechanism called Random Uncorrelated Meta-training (RUM). This mechanism proposes to randomly load uncorrelated batches as support and query sets respectively in the data separation process, then utilize bi-level optimization to directly train the model on visual grounding datasets. Comprehensive experiments on four widely used datasets, as well as in small-sample scenarios, validate the efficacy of MetaVG.**

***Index Terms*—Bi-level optimization, meta-learning, visual grounding.**

I. Introduction

VISUAL grounding is the task of identifying and localizing specific regions or objects within an image using corresponding language expressions. In recent years, significant progress has been made in visual grounding, particularly in the field of visual grounding on remote sensing data [1], [2], [3]. Visual grounding combines language understanding with computer vision to enhance image comprehension and analysis. Additionally, it supports applications such as image captioning [4], cross-modal retrieval [5], [6], [7], [8], and visual question answering [9]. Overall, visual grounding plays a crucial role in advancing the fields of language understanding and computer vision, carrying substantial research significance.

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Recent advancements in visual grounding can be classified into two-stage methods [10], [11], [12], [13], one-stage methods [14], [15], [16], [17], and transformer-based methods [18], [19]. In two-stage methods, the initial stage involves generating region proposals, followed by a subsequent stage where these proposals are matched with language expressions to identify the optimal proposal. One-stage methods directly integrate linguistic context and visual features across all spatial locations to predict bounding boxes without pre-generating object proposals. However, the effectiveness of both two-stage and one-stage methods is constrained by the capabilities of pre-trained structures. To overcome this limitation, transformer-based methods like TransVG [18] and VLTVG [19] have emerged. TransVG adopts a transformer-based approach using multiple transformer encoders. VLTVG focuses on establishing text-conditioned discriminative features and employs multi-stage cross-modal reasoning, achieving state-of-the-art performance on widely used visual grounding datasets.

Although the above-mentioned methods have demonstrated significant effectiveness, they could face challenges when dealing with substantial distribution differences between the training and testing sets. In such cases, these methods tend to overly focus on the training sets, leading to a lack of generalization capability and resulting in overfitting issue. This problem is particularly pronounced in small-sample scenarios. Meta-learning approaches [20], [21] have gained significant popularity in alleviating overfitting, and previous works [22], [23], [24] have demonstrated their effectiveness in utilizing meta-learning to enhance the performance of machine-learning models. Inspired by these methods, we propose a novel meta-learning-based training framework for visual grounding called MetaVG, which directly trains the model on visual grounding datasets. MetaVG demonstrates its efficacy in mitigating overfitting issue without requiring additional auxiliary data or data augmentation, which is typically involved in traditional methods [25], [26], [27]. To effectively train MetaVG directly on the visual grounding datasets, we propose a random uncorrelated meta-training (RUM) mechanism, as illustrated in Fig. 1, which includes the data separation process and the bi-level optimization process. In the data separation process, two uncorrelated batches of data from the datasets, i.e., two batches with no common elements, are loaded as the support and query sets for each episode. In the bi-level optimization process, the model learns priors from the support set within the inner loop and obtains well-performing weight parameters in the outer loop, allowing it to quickly adapt to the target visual grounding task and extract crucial patterns and features from the data, thereby alleviating the overfitting issue.

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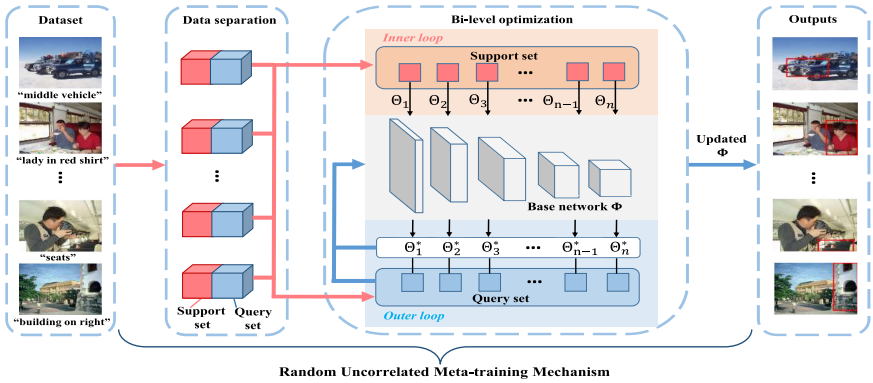


Fig. 1. Illustration of our proposed MetaVG. We propose the random uncorrelated meta-training (RUM) mechanism to construct the episode that is composed of the support set and query set, and then the model is trained and optimized with these episodes in the inner loop and outer loop. Overall, the model learns optimized weights and outputs the target location.

The major contributions of this letter are summarized below:

1) We propose a novel meta-learning training framework, termed MetaVG, to alleviate the overfitting issue in visual grounding. To the best of our knowledge, this work could be the first study on utilizing meta-learning for visual grounding task.
2) We present a novel random uncorrelated meta-training (RUM) mechanism, allowing MetaVG to separate the visual grounding datasets in a random and unrelated manner, and directly conduct bi-level optimization of meta-learning training. This mechanism eliminates the need for additional auxiliary data or data augmentation, which is typically used to alleviate overfitting.
3) Extensive experiments are conducted on four benchmarks, demonstrating the effectiveness of the proposed MetaVG. Notably, MetaVG achieves significant improvement with a large margin in small-sample scenarios compared to the previous state-of-the-art method.

II. Method

A. Overview

For a target visual grounding task, each sample in the training data consists of an image and a corresponding language expression. The image is input into a convolutional neural network with transformer encoder layers to generate a 2D visual feature map. The language expression is encoded as a series of textual embeddings using BERT [28]. Then the features of both modalities are utilized for target localization based on basic network [19]. However, the significant distribution differences between training and testing sets of visual grounding task lead to overfitting. To address this issue, we present a novel MetaVG framework. MetaVG utilizes the proposed random uncorrelated meta-training (RUM) mechanism to first separate two disjoint batches into the support and query sets, and then conducts bi-level optimization with inner-loop optimization using the support set and outer-loop optimization using the query set. We will elaborate on the proposed RUM in the next section.

B. Random Uncorrelated Meta-Training

The support set and query set separation: The support/query episodic training strategy is widely applied in meta-learning [20], [29], [30]. The episode is composed of the support set and query set, which are disjoint. The support set is of great significance for the model to learn a hypothesis. In order to enable meta-learning training using only the original visual grounding datasets, we propose to randomly separate the support set and query set in an uncorrelated way. For the visual grounding task, suppose we have training data $\mathcal{T} = \{T_i\}_{i=1}^{N}$, where $T_i = (S_i, Q_i)$ is a training episode with support set S_i and query set Q_i. And N is the number of episodes. We randomly select two uncorrelated batches from the visual grounding datasets as the support set S_i and the query set Q_i for an episode T_i respectively. Note that S_i and Q_i are uncorrelated because the training samples in these two batches are disjoint, i.e., $S_i \cap Q_i = \emptyset$. For support set $S_i = \{X_k, Y_k\}_{k=1}^{N_B}$, each pair X_k and Y_k represents a sample pair of the input image with its corresponding language expression and the output phrase-box; N_B is the batch size. The sample pair format of the query set is the same as the support set. In this way, we successfully separate the required support set and query set, laying the data foundation for the bi-level optimization [31] of our RUM mechanism.

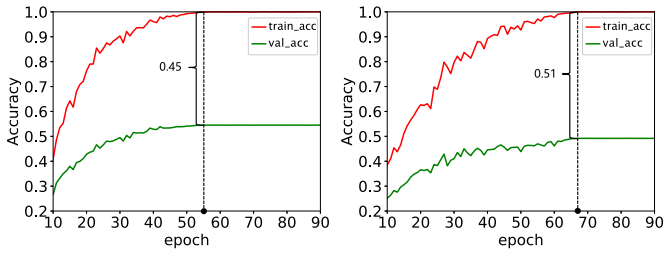
Inner-loop optimization: To enable the model to learn the hypothesis so that it can quickly adapt to the target visual grounding task, we train the model on the support set and this process is the inner loop of bi-level optimization. Denote the weight parameters of the model as $\mathbf{\Phi}$, which are initialized randomly. The parameters for training episodes can be denoted as $\{\mathbf{\Theta_i}\}_{i=1}^{N}$. Initially, we set $\mathbf{\Theta_1} = \mathbf{\Phi}$. When adapting to the support set S_i of episode T_i, the parameters $\mathbf{\Theta_i}$ are updated to $\mathbf{\Theta_i^*}$. With a loss function $\mathcal{L}|_{(S_i,\mathbf{\Theta_i})}$, we use gradient descent to compute the updated $\mathbf{\Theta_i^*}$ for episode T_i:

$$\mathbf{\Theta_i^*} = \mathbf{\Theta_i} - \alpha \frac{\partial \mathcal{L}|_{(S_i,\mathbf{\Theta_i})}}{\partial \mathbf{\Theta_i}}, \tag{1}$$

where α is a hyperparameter that denotes the learning rate in the inner loop. $\mathbf{\Theta_i^*}$ is the learned hypothesis on episode T_i and the performance of $\mathbf{\Theta_i^*}$ will be measured on the corresponding query set Q_i in the outer loop.

Outer-loop optimization: During the outer-loop optimization, the model takes $\mathbf{\Theta_i^*}$ and the query set Q_i as inputs to evaluate the performance of learned hypothesis. Then the model updates the weight parameters $\mathbf{\Phi}$ by performing episode-level stochastic gradient descent (SGD). Based on this, we propose to utilize

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(a) MetaVG(Ours) (b) VLTVG

Fig. 2. The average learning curves of MetaVG. (a) VLTVG. (b) In the 1000-shot scenario on the RefCOCO dataset are obtained through five training rounds, each utilizing different random seeds. We plot the accuracy of the training (red) and validation (green) process.

a meta-learning objective for the target visual grounding task, which aims to guide the optimization of $\mathbf{\Phi}$. Specifically, based on (1), the objective is written below:

$$\min_{\mathbf{\Phi}} \sum_{i=1}^{N} \mathcal{L}|_{(Q_i, \mathbf{\Theta}_i^*)}, \tag{2}$$

where N denotes the number of training episodes. Note that our goal is to obtain well-performing weight parameters $\mathbf{\Phi}$ based on hypothesis, therefore the computing of the meta-learning objective is based on parameters $\{\mathbf{\Theta}_i^*\}_{i=1}^N$ obtained in the inner loop.

We use SGD to solve (2) involving computing gradient of a gradient for the first term that can be derived as:

$$\Delta(\mathbf{\Theta}_i^*; Q_i) = \frac{\partial \mathcal{L}|_{(Q_i, \mathbf{\Theta}_i^*)}}{\partial \mathbf{\Theta}_i^*} \left(1 - \alpha \frac{\partial^2 \mathcal{L}|_{(S_i, \mathbf{\Phi})}}{\partial \mathbf{\Phi}^2}\right). \tag{3}$$

Based on (3), the weight parameters $\mathbf{\Phi}$ can be updated as:

$$\mathbf{\Phi} \leftarrow \mathbf{\Phi} - \beta \frac{1}{N} \sum_{i=1}^{N} \Delta\left(\mathbf{\Theta}_i^*; Q_i\right), \tag{4}$$

where β is the learning rate in the outer loop. With the bi-level optimization, the network parameters could converge to a solution that alleviates overfitting, as shown in Fig. 2.

C. Training Loss

Since the network outputs the prediction phrase-box as a 4-dimensional vector [19], we can compute the regression loss directly with the ground-truth box, which simplifies positive/negative examples mining in training. We follow the previous work [19] denoting the ground-truth box as b, and the prediction bounding boxes from stages 1 to N_s as $\{\hat{b_i}\}_{i=1}^{N_s}$, such that the training loss can be written as:

$$\mathcal{L} = \sum_{i=1}^{N_s} \left(\lambda_{L1} \mathcal{L}_{L1}\left(b, \hat{b_i}\right) + \lambda_{giou} \mathcal{L}_{giou}\left(b, \hat{b_i}\right)\right), \tag{5}$$

where $\mathcal{L}_{L1}(\cdot,\cdot)$ and $\mathcal{L}_{giou}(\cdot,\cdot)$ are the L1 loss and GIoU loss [32] respectively. The λ_{L1} and λ_{giou} are the hyperparameters to balance these two losses.

III. Experiment

A. Datasets

ReferItGame: ReferItGame [33] is divided into three subsets, namely training, validation, and testing sets, with 54,127, 5,842, and 60,103 referring expressions respectively, which have the same split with the previous works [16], [34].

RefCOCO/RefCOCO+/RefCOCOg: RefCOCO [35], RefCOCO+ [35] and RefCOCOg [36] are splits of the MS COCO dataset [37]. RefCOCO consists of 120,624, 10,834, 5,657, and 5,095 expressions for train, validation, testA, and testB, respectively. Similarly, RefCOCO+ has 120,191, 10,785, 5,726, and 4,889 referring expressions for the corresponding subsets. RefCOCOg is composed of 25,799 images, which contain 95,010 referring expressions for 49,822 referents. The dataset is divided into two subsets, RefCOCOg-google [36] (val-g) and RefCOCOg-umd [38] (val-u, test-u).

B. Implementation Details

Our CNN backbone is ResNet-50, which is followed by six transformer encoder layers. These layers are initialized with the weights from the DETR model [39], which is also adopted in the work [19]. The linguistic module is initialized with BERT [28]. Following [19], $\lambda_{giou} = 2$ and $\lambda_{L1} = 5$ for the loss function in (5).

For the inner-loop optimization during meta-learning, we use the AdamW optimizer [40] with the learning rate α of 1e-4 for the first 60 epochs, and subsequently maintain it at 1e-5. For the outer-loop optimization in meta-learning, we apply the SGD optimizer with a learning rate β of 1e-5. To ensure training stability, we keep the weights of the visual and textual feature extraction branches frozen for the initial 10 epochs. When comparing with state-of-the-art method in the context of limited data samples, we set the number of training epochs for the model to be 90. We train our MetaVG with Pytorch interface on Tesla V100. The batch size for all experiments is set to 8. Following the methodology of prior studies [16], [19], we adopt the train, validation, testA, and testB splits for RefCOCO and RefCOCO+. The images in the "testA" subset depict multiple people, while the images in the "testB" subset encompass various other objects.

The evaluation protocol is the same as prior works [18], [19]. When the IoU between the predicted region and the ground-truth box exceeds 0.5, the prediction is deemed correct.

C. Comparisons With State-of-The-Art Methods

We report our performance and compare it with other state-of-the-art methods on four benchmark visual base datasets. Table I reports visual grounding results on ReferItGame, RefCOCO, RefCOCO+, and RefCOCOg. With the ResNet-50 backbone, MetaVG achieves 74.73% accuracy and outperforms the state-of-the-art method VLTVG with the same backbone network by 3.13% on the dataset ReferItGame. Our accuracies exceed VLTVG by 0.23% $\sim$ 0.94% on RefCOCO, 0.40% $\sim$ 1.12% on RefCOCO+, and 0.28% $\sim$ 0.65% on RefCOCOg. As shown in Table I, our accuracies also surpass the previous two-stage and one-stage methods in all subsets of the aforementioned four datasets. These results demonstrate our state-of-the-art performance on four widely used datasets.

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TABLE I
THE ACCURACY(%) COMPARISONS ON REFERITGAME, REFCOCO, REFCOCO+ AND REFCOCOG

	Method	Backbone	ReferItGame	RefCOCO			RefCOCO+			RefCOCOg		
				val	testA	testB	val	testA	testB	val-g	val-u	test-u
Two-stage	CMN [10]	VGG16	28.33	-	71.03	65.77	-	54.32	47.76	57.47	-	-
	MAttNet [11]	ResNet-101	29.04	76.65	81.14	69.99	65.33	71.62	56.02	-	66.58	67.27
	NMTree [12]	ResNet-101	-	76.41	81.21	70.09	66.46	72.02	57.52	64.62	65.87	66.44
	Ref-NMS [13]	ResNet-101	-	80.70	84.00	76.04	68.25	73.68	59.42	-	70.55	70.62
One-stage	FAOA [14]	DarkNet-53	60.67	72.54	74.35	68.50	56.81	60.23	49.60	56.12	61.33	60.36
	RCCF [15]	DLA-34	63.79	-	81.06	71.85	-	70.35	56.32	-	-	65.73
	ReSC-Large [16]	DarkNet-53	64.60	77.63	80.45	72.30	63.59	68.36	56.81	63.12	67.30	67.20
	LBYL-Net [17]	DarkNet-53	67.47	79.67	82.91	74.15	68.64	73.38	59.49	62.70	-	-
T-based	TransVG [18]	ResNet-50	69.76	80.32	82.67	78.12	63.50	68.15	55.63	66.56	67.66	67.44
	VLTVG [19]	ResNet-50	71.60	84.53	87.69	79.22	73.60	78.37	64.53	72.53	74.90	73.88
	MetaVG(Ours)	ResNet-50	**74.73**	**84.76**	**88.07**	**80.16**	**74.12**	**78.77**	**65.65**	**73.18**	**75.18**	**74.20**

We highlight the best and the second best performance with bold and underline, respectively. "T-based" refers to "transformer-based".

TABLE II
THE MEAN ACCURACY(%) WITH STANDARD DEVIATION OF METAVG IN SMALL-SAMPLE SCENARIOS COMPARED WITH VLTVG

Datasets	Methods	1000-shot	500-shot	100-shot	20-shot
RefCOCO	VLTVG	49.19±1.99	31.66±1.43	14.41±2.32	8.27±1.62
	MetaVG	**54.51±0.17**	**42.51±0.12**	**23.57±0.28**	**9.83±0.19**
RefCOCO+	VLTVG	46.36±1.65	31.72±2.67	13.57±2.78	8.73±1.77
	MetaVG	**49.63±0.25**	**36.61±0.53**	**16.75±0.21**	**10.38±0.35**
ReferItGame	VLTVG	30.62±2.52	25.08±2.41	11.17±1.82	3.36±1.71
	MetaVG	**37.36±0.61**	**26.45±0.34**	**15.51±0.25**	**5.56±0.38**

We highlight the best performance with bold.

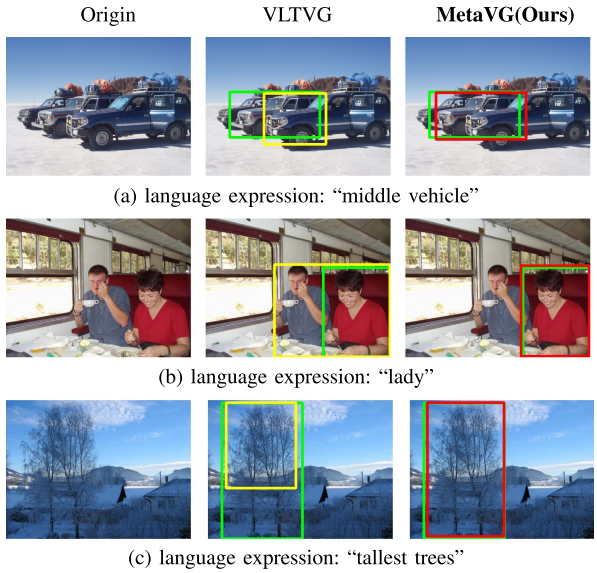
D. Comparison With VLTVG in Small-Sample Scenarios

To verify the effectiveness of MetaVG in mitigating overfitting in small-sample scenarios, we conduct a comparative performance analysis between MetaVG and the state-of-the-art method VLTVG. We randomly extract 1000, 500, 100, and 20 samples five times from the training sets of RefCOCO, RefCOCO+, and ReferItGame, respectively, to create small-sample settings. Additionally, the configuration of the test sets remains consistent with the three datasets. The utilization of the RUM mechanism is restricted to the training phase, and the test phase follows the standard visual grounding testing phase.

In all small-sample settings, our MetaVG surpasses VLTVG by a noticeable margin, as shown in Table II. Notably, MetaVG achieves significant improvements in the 100-shot scenario, with mean accuracies exceeding VLTVG by 9.16%, 3.18%, and 4.34% in RefCOCO, RefCOCO+, and ReferItGame datasets respectively. Fig. 2 demonstrates that MetaVG exhibits greater stability and faster convergence compared to VLTVG during training. MetaVG converges in epoch 55 while VLTVG converges in epoch 67. Additionally, MetaVG consistently achieves higher accuracy while exhibiting a smaller standard deviation compared to VLTVG. These results demonstrate the efficacy of our proposed MetaVG framework in alleviating overfitting in small-sample visual grounding scenarios.

E. Visualization

To demonstrate the superiority of MetaVG, we provide visual examples in Fig. 3. The left, middle, and right columns show the original images, VLTVG's results, and MetaVG's results, respectively. One could observe that MetaVG captures finer features, even in challenging scenarios. For example, in Fig. 3(a) with the expression "middle vehicle", MetaVG accurately frames the obscured middle vehicle, while VLTVG only captures the rear half. Thanks to our RUM mechanism used in MetaVG, our model fully exploits latent features and focuses on more accurate text semantics.



(a) language expression: "middle vehicle"

(b) language expression: "lady"

(c) language expression: "tallest trees"

Fig. 3. Visualization of examples. Predicted regions of VLTVG, MetaVG, and the ground truth are boxed in yellow, red, and green, *resp.*

IV. CONCLUSION

In this letter, we present a new meta-learning training framework for visual grounding, termed MetaVG, which alleviates the overfitting problem caused by distribution differences of the datasets. To the best of our knowledge, this work could be the first attempt at utilizing meta-learning for visual grounding. Moreover, we propose a novel random uncorrelated meta-training (RUM) mechanism that aims to construct the support and query set for bi-level optimization of meta-training so that MetaVG can be directly trained on the visual grounding datasets. Experiments on four widely used datasets and in small-sample scenarios validate the efficacy of our approach.

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